

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Mock Paper Set 4

Paper reference	8MA0/21
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Mathematics
Advanced Subsidiary
PAPER 21: Statistics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

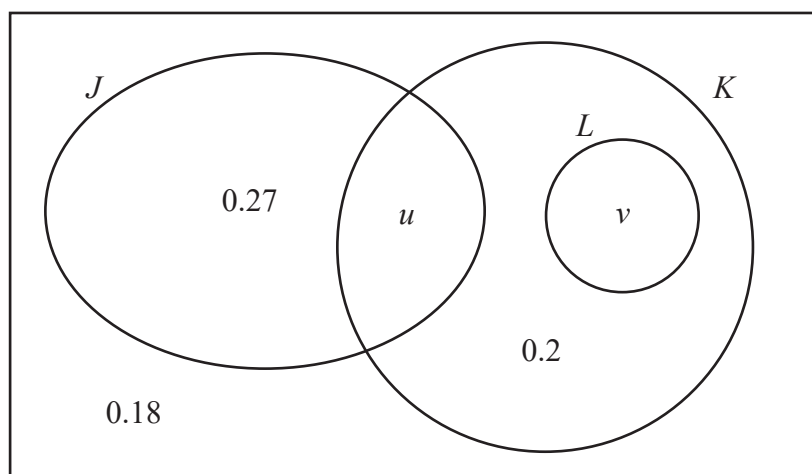
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1. The Venn diagram represents three events J , K and L and their associated probabilities. Two of the associated probabilities are u and v



- (a) State which two events from J , K and L are mutually exclusive. (1)
- (b) Show that $u + v = 0.35$ (1)
- The events J and K are independent.
- (c) (i) Find an equation in terms of u (2)
- (ii) Hence find the value of u and the value of v (2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. A restaurant has only two options, a meat option (menu A) and a meat-free option (menu B).

The restaurant manager knows that 29% of customers choose menu B.

The manager decides to use a binomial distribution to model the number of customers choosing menu B.

The restaurant serves 40 customers each evening.

- (a) Using the manager's model, show that, on a randomly selected evening, the probability that more than a quarter of the customers choose menu B is 0.641 to 3 significant figures.

(2)

The restaurant is open 5 evenings each week.

- (b) (i) Find the probability that in a randomly selected week, more than a quarter of customers choose menu B on one evening or fewer.

(3)

- (ii) Give one assumption necessary for the answer to part (b)(i) to be valid.

(1)

The manager decides to change menu B from meat-free to vegan.

On a randomly selected evening after this change, 6 out of the 40 customers chose menu B.

- (c) Using a suitable hypothesis test, investigate whether the proportion of customers choosing menu B has changed.

You should use a 5% level of significance and state your hypotheses clearly.

(4)



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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)



3. Joe wants to compare Daily Mean Total Cloud Cover (oktas), on any given day, in Leeming and Camborne.

Joe takes a sample of the 2015 data from the large data set for each of these two locations.

- (a) (i) Describe how a **systematic** sample of 30 cloud cover readings could be taken from the large data set for Leeming in 2015. (2)
- (ii) Explain briefly how the sample for Camborne should now be taken. (1)

Joe believes that locations further from the coast have greater levels of cloud cover than locations nearer to the coast.

Joe's data shows a mean cloud cover of

- 5.0 oktas for Leeming
- 5.4 oktas for Camborne

- (b) State, giving a reason, whether these statistics are consistent with Joe's belief. (2)

Joe wishes to model cloud cover, in oktas, as a discrete random variable.

- (c) State (2)
- a possible distribution Joe could use for this model
 - an assumption required for this distribution to be suitable



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



- The sample is summarised by

(a) Find the variance of the daily yields of the machines in factory A.

The sample is summarised by

(b) Find the combined variance of the daily yields of the machines in factory A and factory B.

(5)

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Question 4 continued

Lined area for writing the answer to Question 4.



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Question 4 continued

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(Total for Question 4 is 7 marks)

TOTAL FOR STATISTICS IS 30 MARKS

